

Electrical Powertrain & System Report for Electric-All Terrain Vehicle

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1. Introduction

Team Aveon Racing is a passionate and hardworking student team made up of engineering students from various branches of engineering, including Mechanical, Electrical & Electronics, Electronics & Communication, and Computer Science. We work together to design, build, and test an Electric All-Terrain Vehicle (e-ATV) to compete in the national-level event , e-BAJA SAE INDIA.

Our team was started with a clear goal to turn classroom knowledge into hands-on innovation and promote research in the field of electric vehicles. We are proud to be the first e-BAJA team from East India, and we aim to set a high standard not just through our technical work, but also by building strong teamwork and a shared passion for high-performance electric vehicles.

The e-BAJA SAE INDIA competition challenges teams from across the country to design and manufacture a tough, single-seater e-ATV that can perform in extreme conditions including a series of performance tests and a final endurance race. This is more than just a competition for us; it's a chance to apply what we've learned, grow as engineers, and push the boundaries of electric mobility.

This report gives a detailed look into the electrical powertrain and electrical system of our e-atv which we made for the 2025 e-BAJA SAE INDIA competition .

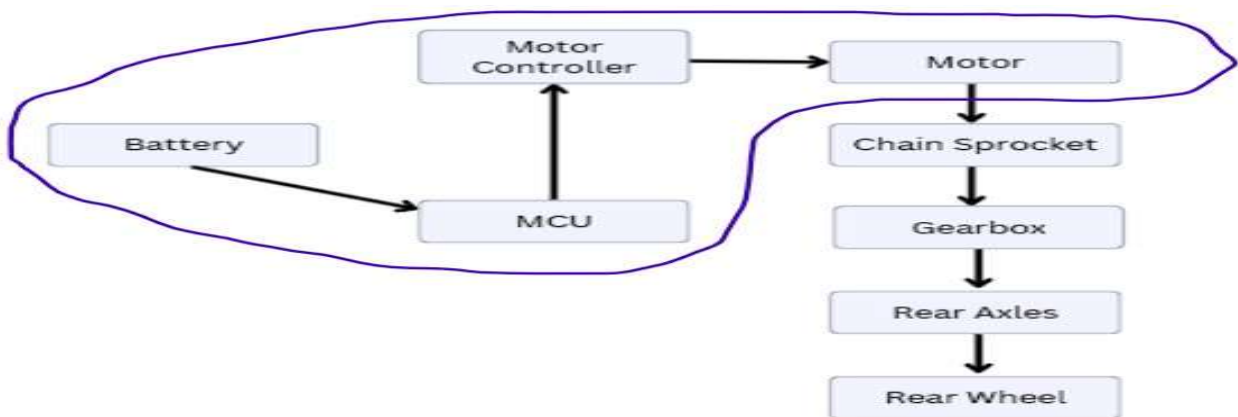


Fig1. Position of Electrical Powertrain in the Overall Power Flow

2. Electrical System

The electrical system of the EV consists of 2 parts, a high voltage tractive system (48 volts) and a low voltage auxiliary system (12 volts). The tractive system connects the battery to the motor controller (which drives the PMSM motor) and buck converter which reduces the 48 volts to 12 volts for operation of the LV system. The low voltage system consists of TSAL, RTDS , reverse and brake lights and reverse alarm. The isolation of the tractive system is done by the use of 2 Accumulator Isolation Relays positioned on both terminals of the battery. The battery management system is integrated with the battery. The switches to operate low voltage and high voltage systems are located on the dashboard. The foot pedal consists of a potentiometer which sends data to the motor controller directly.

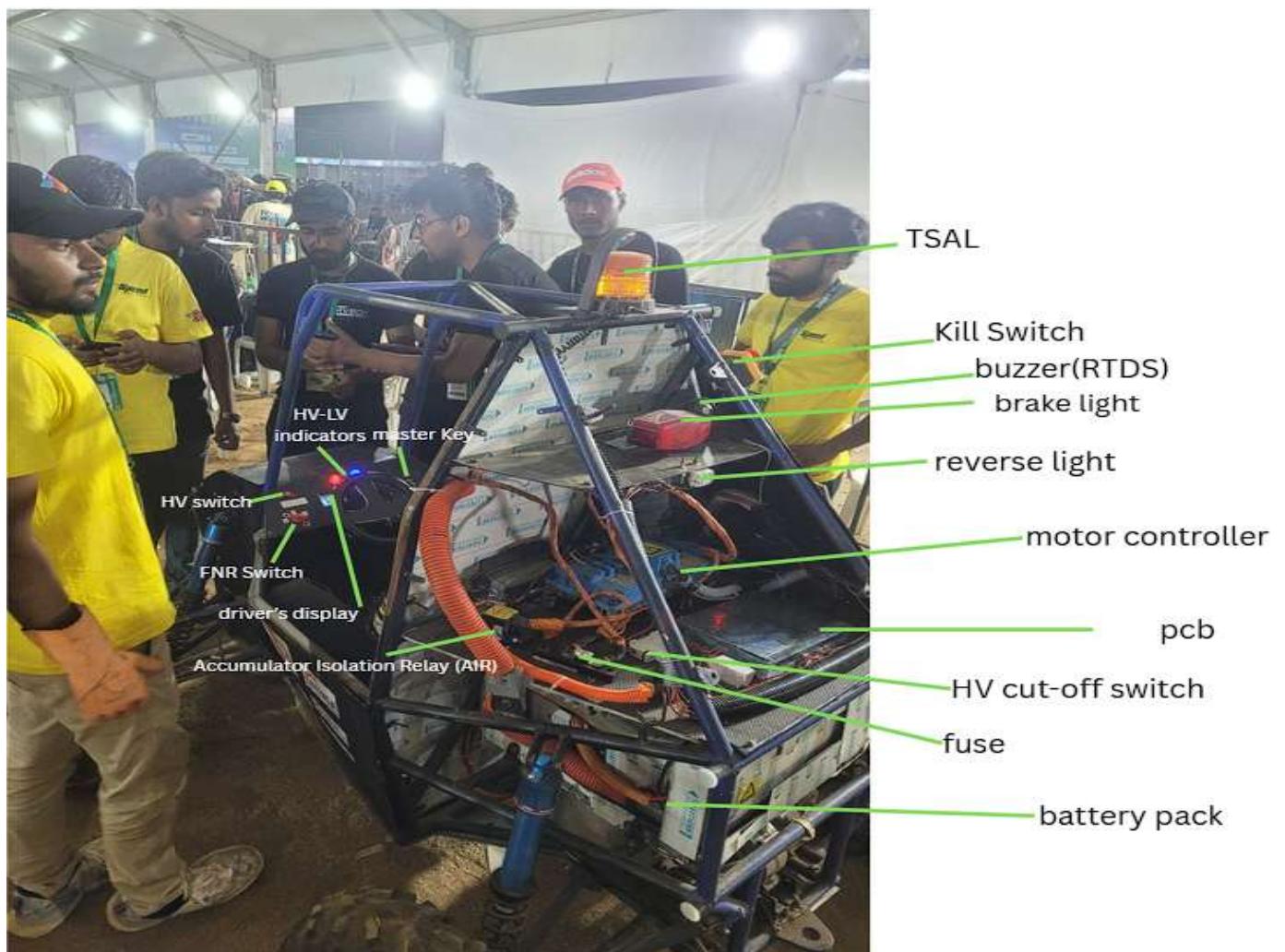
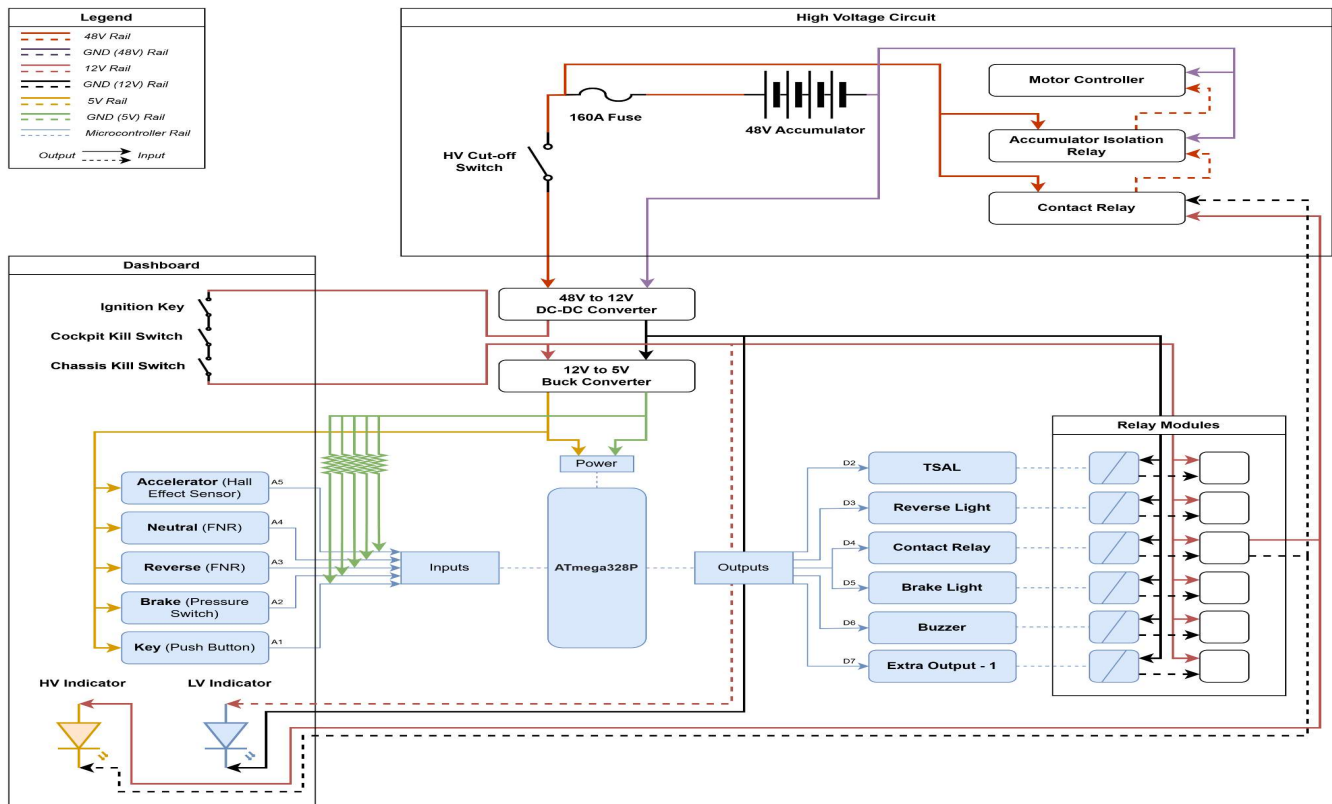


Fig2. components of the electrical system of our e-atv (2025)

❖ Block Diagram of Overall Electrical System (HV and LV system) of the vehicle



3. High Voltage Traction System

It deals with High Voltage system components namely HV Battery and Electric Motor. The tractive system of the vehicle is defined as every part that is electrically connected to the motor and tractive system Accumulator. The Voltage between any two terminals of the tractive system in the vehicle shall not exceed 60V DC in any condition (according to the rulebook). All components in the tractive system must be rated for the maximum tractive system voltage. The tractive system Accumulator is defined as all the battery cells that store the electrical energy to be used by the tractive system along with the corresponding contactor, fuse, and battery management systems.

The tractive system motor must be connected to the Accumulator(s) through a motor controller. Bypassing the control system and connecting the tractive batteries directly to the motor is strictly prohibited.

3.1 Battery Pack

The energy required to power the vehicle is stored in the battery pack, which is made up of many cells. Direct current output is provided by a battery pack. The bulk of portable consumer devices currently employ lithium-ion batteries due to their high energy density per mass compared to other electrical energy storage solutions. They also work well at high temperatures, have a high power-to-weight ratio, are highly energy-efficient, and discharge themselves slowly. The battery pack is bought from Adiabatic Technologies Private limited . It is of NMC chemistry type and has 48 V 120 Ah capacity

The battery pack is mounted on the area above the S-R plane such that the path of battery removal during battery swapping should be easier and accessible both in horizontal and vertical direction of movement.

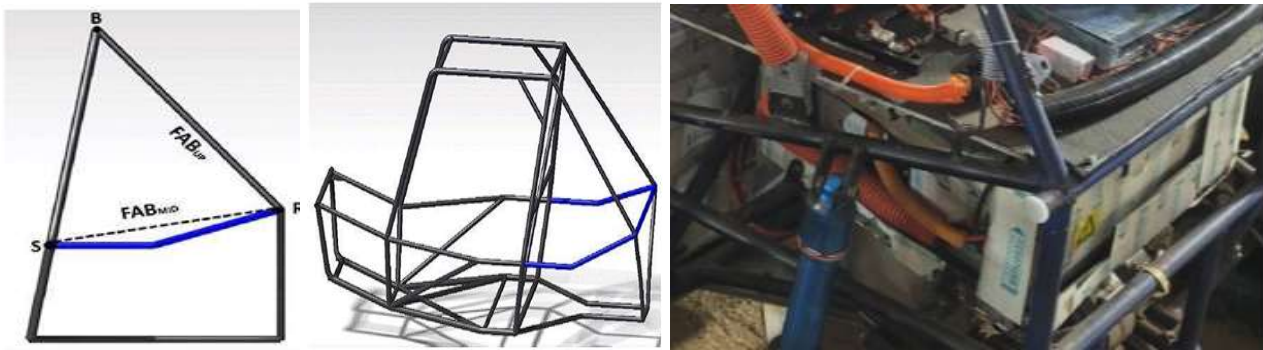


Fig3. battery pack mounting area

ELECTRICAL SPECIFICATIONS		CHARGE SPECIFICATIONS	
Nominal Voltage	48V	Recommended Charge Current	15A - 80A
Nominal Capacity	100AH	Discharge Temperature	-4°C to 60°C
Bluetooth Enabled\IOT Enabled	Yes	Charge Temperature	-4°C to 55°C
Efficiency	99%	Recommended Charge Voltage	37.5V - 54.6V
Capacity @ 15 A	99%	BMS Charge Voltage Cut off	54.6V (4.2±0.05Vpc)
Self Discharge	3% /month	Reconnect Voltage	53.3V (4.1±0.05Vpc)
Maximum Modules in Parallel and Series	15S1P		

DISCHARGE SPECIFICATIONS		MECHANICAL SPECIFICATIONS	
Max Continuous Discharge Current	120A	Dimension (MM)	366x312x180(mm)
Peak Discharge Current	200A(10S)	Weight	42Kg
BMS Discharge Current Cutoff	210A±2A; 1S	Terminal Type	IP67
Low voltage Disconnect	41.25V(2.75V)	Terminal	SB120
BMS Discharge Voltage Cut off	37.5V(2.5V)	Case Material	MS Powder Coat
Reconnect Voltage	43V	Enclosure Protection	IP67
Short Circuit Protection	1ms	Cell Type-Chemistry	NMC
		Life Cycles	3000

Fig4. specifications of the battery pack

3.2 Battery Management System

The Daly BMS is fixed as OEM and is fitted inside the battery pack enclosure itself. The BMS satisfies all the standards specified in the rule book. The battery BMS data is analysed using the smartBMS app which communicates with the BMS using Bluetooth communication.

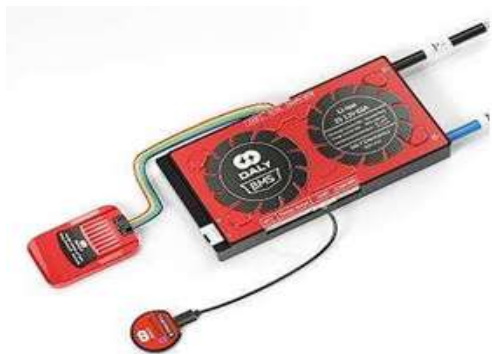


Fig5. Battery Management System

3.3 Charger

The Charger is also bought from Adiabatic Technologies Private Limited . The charging environment at the garage is implemented as per charging safety standards mentioned in rulebook.

3.4 Electric Motor

The motor is chosen as Permanent Magnet Synchronous Motor (PMSM) for better controlling and traction on any type of surface. It is bought from Datai corporation. It has 6 phases and the coils are energized according to hall sensor feedback signal and throttle input signal

	Motor		
	Make / Model	DATAI	
	Operating Voltage (V)	48V	
	Cont. Rated Power (kW) @ RPM	5kW	@ 3600 RPM
	Max Peak Power (kW) @ RPM	7.2kW	@ 2700 RPM
	Rated Speed (RPM)	3900 RPM	
	Maximum Speed (RPM)	4500 RPM	
	Cont. Rated Torque (Nm) @ RPM	25Nm	@ 2700RPM
	Max Peak Torque (Nm) @ RPM	75Nm	@ 300RPM
	Cooling Type	AIR COOLING	
	Cooling system is powered of	connected with motor shaft	
	Age of the Motor if reused (in Years)	0	
	Operating Temperature range (°C)	0-50	
	Vendor details	Eifer Megacorp Pvt Ltd	

Fig6. Electric Motor and its specification

3.5 Motor Controller

A motor controller is a piece of equipment, or group of equipment, that has the ability to regulate how an electric motor operates in accordance with established rules. A motor controller makes it possible to start and stop the motor automatically or manually, select forward or reverse rotation, choose and regulate the speed, restrict or control the torque, and prevent overloads and electrical failures. The Motor Controller used is bought from DATAI corporation .

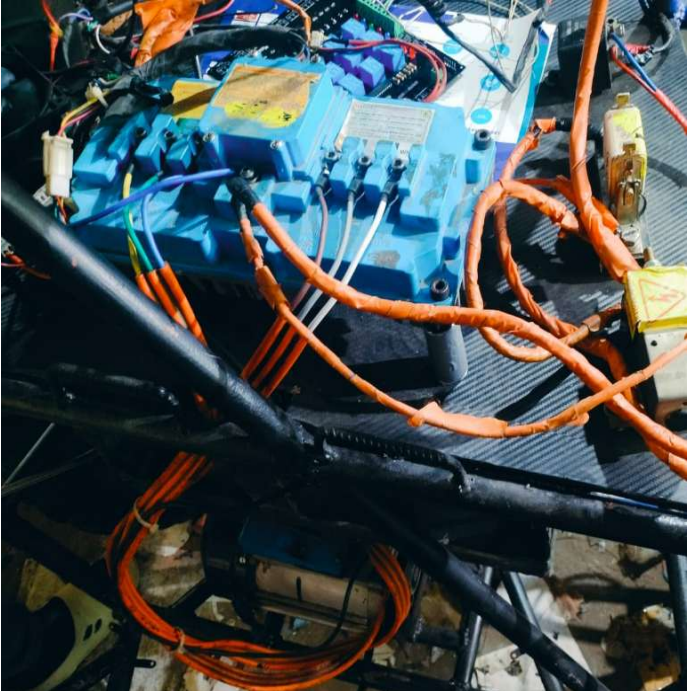
		Electric Motor Controller	
Make / Model		DATAI	
Available Battery Voltage (V)		48	
Available Motor Power (kW)		5	
Peak Phase Current (A)		25	
Rated DC Current (A)		120	
Maximum DC Current (A)		150	
Communication Protocol		Not applicable	
Throttle Input Type		Pedal feedback	
Operating Temperature Range		0 °C – 50 °C	
Vendor details		Eifer Megacorp Pvt Ltd	

Fig7. Motor Controller and it's specification

3.6 DC-DC Converter

A direct current (DC) to direct current (DC) converter is a circuit or electrical device that alters the voltage level of a direct current (DC) source. It's a type of electromagnetic power converter. Tiny batteries have very little power, whereas high-voltage power transfer has a lot. DC-to-DC converter circuitry is often used to adjust the output voltage. The DC-DC converter is bought from Traco Power . It reduces the accumulator voltage (48V) Into 12V which is then fed up to the PCB.



4.1 DC-DC Converter	
Type	TEQ 300WIR Series
Input rated voltage (V)	48
Input voltage range (V)	18 - 75
Efficiency (%)	89%
Output voltage (V)	12
Output current (A)	25
Output rated power (W)	300
Output peak power (W)	345
Voltage regulation (%)	0.2 max
Load regulation (%)	0.5 max
Ripple (full load test)	125 mVp-p max
No-load current (A)	0.03
Working Temperature (°C)	-40 to +80
IP Rating	IP67
Protections (Over-volt, Under-volt etc.,)	Over-volt, Under-volt, Over-temp
Fuse rating (A)	25

Fig17. DC-DC Converter and it's specification

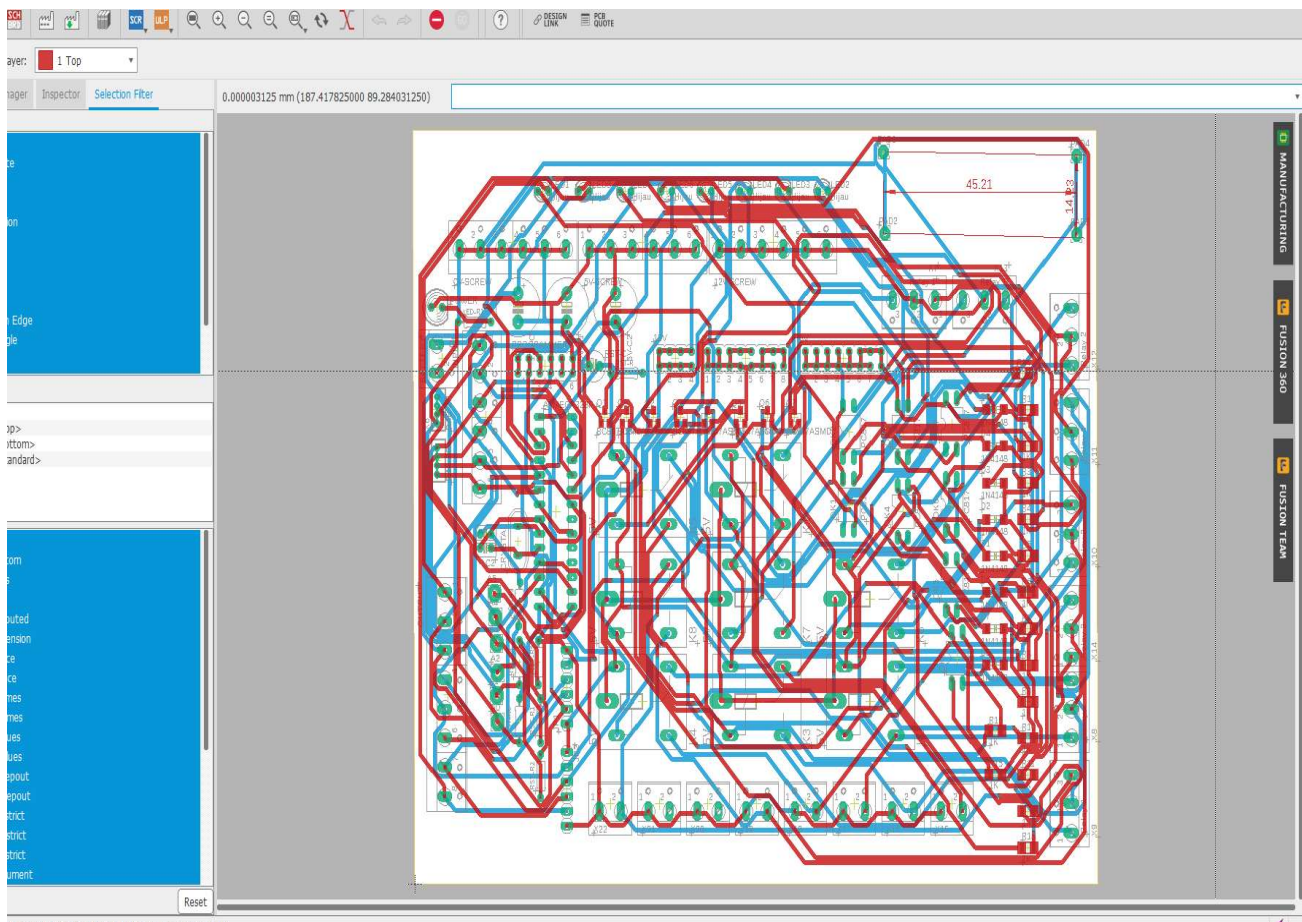
4. Main Control Unit (MCU)

The main control unit provides logical control and implementation of all the starting and security sequences according to the rulebook . The drive mode operates whenever the brake pedal is pressed, neutral mode is engaged, and the accelerator pedal is not depressed. The control panel, a critical component of this system, is constructed using normally open (NO) and normally closed (NC) relays,

We have developed a custom-designed Printed Circuit Board (PCB) that incorporates comprehensive design considerations, including robust isolation measures and enhanced safety features through the strategic use of relays, transistors, and opto-isolators. This PCB receives its primary power from a 12V DC-DC converter, with an integrated 5V buck converter on the board providing the necessary 5V power supply for its internal components.

❖ PCB Designing

The PCB is designed on EAGLE Software and the design is manufactured by getting gerber files to a manufacturer



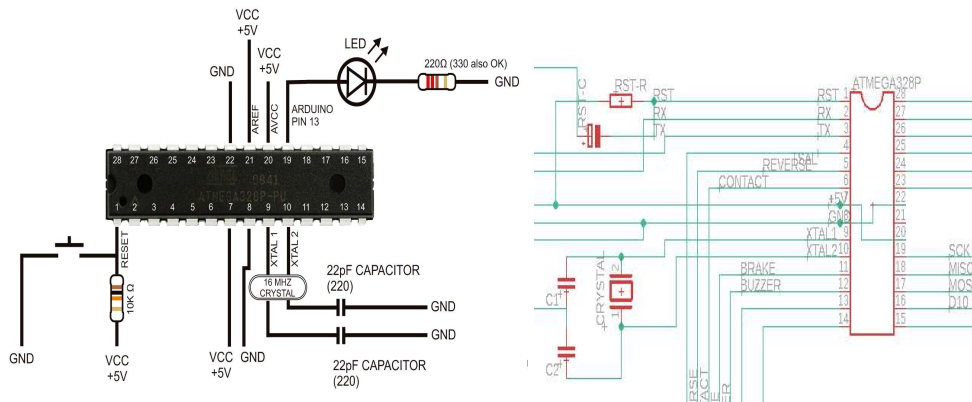


Fig10. ATMEGA328P Microcontroller and its basic working circuit

The ATmega328P requires a 5V power supply connected to its VCC and AVCC pins, and a GND connection. The 16 MHz crystal oscillator, along with two 22 pF capacitors, provides a stable clock signal essential for timing and operation. The RESET circuit ensures stable startup and manual reset capability.

This basic circuit which is used to run the ATMEGA microcontroller is used in our pcb and the rest of the vehicle's logic and control of LV system is custom designed using the I/O ports of the microcontroller

The power to this control PCB is taken from the vehicle's main accumulator, which outputs 48V DC.

The microcontroller and signal electronics operate at much lower voltages so this 48V is first stepped down using a DC-DC converter, converting it to 12V DC and then This 12V DC is then fed into a buck converter, which reduces it further to 5V DC. This regulated 5V is used to power all the logic-level components in the circuit, including the microcontroller, optocoupler, LEDs, and relay coils.

Control Circuit :

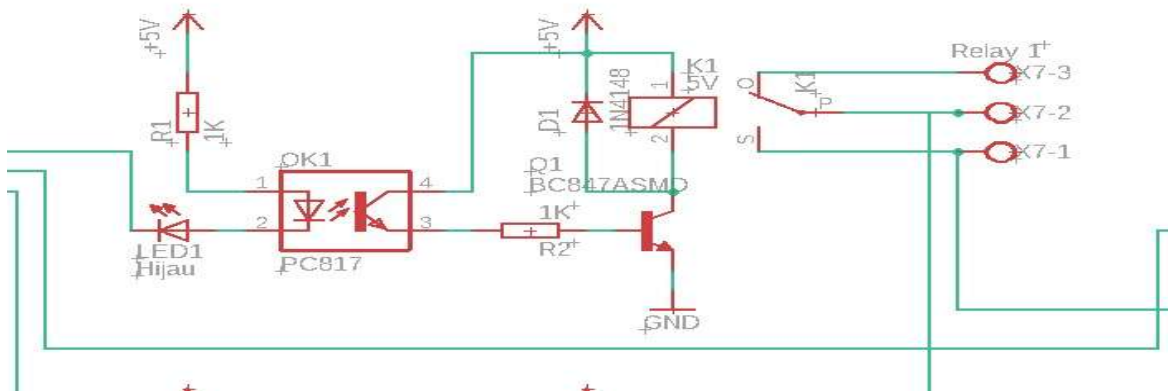


Fig11. Control Circuit Schematic (for one specific control)

The given control circuit is designed for electrical isolation and safe switching of high-power components using low-power control signals, which is crucial for safety in our vehicle . The PC817 optocoupler (OK1) provides isolation between the low-voltage control logic and the high-power side. When a signal is applied to the LED inside the optocoupler, it emits light that activates the phototransistor on the output side, effectively transferring the signal without electrical contact, ensuring safety and protecting sensitive control electronics.

The signal from the optocoupler drives a BC817 NPN transistor (Q1), which acts as a switch. When the transistor is turned on, current flows through the relay coil (k1), energizing it and closing the contacts to control a higher-power load . The relay allows a low-power circuit to control large currents needed for components like motor controller , AIR, Lights of the our atv.

To protect the transistor from voltage spikes caused by the relay coil's inductance during switching, a flyback diode (D1 – 1N4148) is placed across the relay coil. This diode absorbs the back EMF, preventing damage to Q1. The circuit also includes R1 and LED1. R1 limits current through the input LED of the optocoupler, and LED1 is used as a visual indicator that the control signal is active.

This setup ensures isolation, signal amplification, and protection, which are all critical in EV electronics where both safety and noise immunity are key.

To ensure durability and safety during high-vibration and off-road operation, the entire control circuit is housed in a custom-designed PCB box made from ULV90-rated material, as per the competition rulebook. ULV90 is a flame-retardant and impact-resistant material, ideal for electrical enclosures in automotive and rugged applications. All internal wirings are

properly insulated using high-grade insulating sleeves, heat shrink tubing, and organized routing. This prevents any accidental shorts or wire displacement when the ATV is operating on rough terrain, ensuring both safety and long-term reliability of the electrical system.

5. Grounded Low Voltage Systems(GLV)

The GLV system consists of lights, alarms and other ancillaries. The Tractive System Active light (TSAL) indicates the operation of the High Voltage Traction system. The reverse light and reverse alarm indicate the working of reverse operation. The brake light operates whenever the brake is applied. The driver indication lamps indicate the GLV and HV status to the driver. The ignition key is used to start the vehicle. The FNR switch is used to toggle between Forward Drive /Reverse mode /Neutral condition. The kill switches are used to open the complete circuit of the vehicle whenever an emergency situation occurs. The Ready to Drive sound operates whenever the vehicle is started, for 3 seconds.

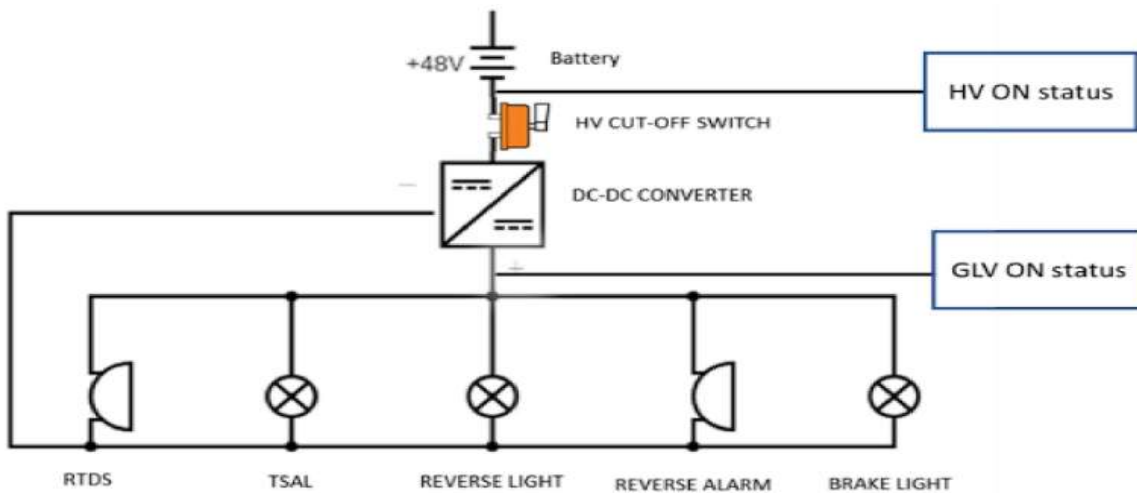


Fig13. GLV System Overview

5.1 Brake Light

The brake light is purchased satisfying the rulebook constraints. AIS standard homologation mark engraved brake light is purchased from authorized dealers. The light is energized whenever the brake is pushed. The brake light energizing circuit simply has two automotive pressure switches, brake light, relay, power source (DC-DC converter). The light is in adherence to BAJA SAE J586 and J759 standards. Hydraulic pressure switches are mounted on the ancillary output ports of the master cylinder. They get activated using the brake fluid pressure during braking conditions and act as a switch closing the brake light circuit.

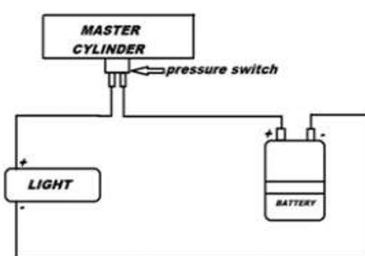


Fig14. Brake circuit , Brake light used and brake pressure switch

5.2 KILL SWITCHES

The kill switches are used to completely cut-off the powertrain circuit from the power source in case of emergency situations and casualties. They de-energize all the components except the brake lights. The kill switch is of E-stop, PUSH TO KILL and PULL TO ROTATE type. Two kill switches are mounted on the vehicle. One is at the Driver cockpit side along the SIM (Side Impact Member) and another one is at the lateral side of the vehicle. The rear kill switch is positioned on the right side of the vehicle, aft of the plane of the RRH, and forward of the right FABUP for easy access for track workers or volunteers.

5.3 REVERSE ALARM

The reverse light and reverse alarm are bought as OEM with AIS standard as specified in rulebook. The reverse light and reverse alarm are energized whenever the D-N-R switch is put on R – Reverse mode. The light has “R” engraving and satisfies SAE J759 standard. The reverse alarm has J17941 standard. The reverse alarm and reverse light are positioned as per rulebook (27.6 inches above ground).



Fig15. Reverse alarm

5.4 TSAL

TSAL (tractive system activated light) is used to indicate the Tractive system ON status. The TSAL is IP65 rated and has >350 Lumens of brightness. It is mounted inside a cage fitted on the centre of BLC (Overhead Lateral Cross member).



Fig16. TSAL and TSAL mounting on vehicle

5.5 IGNITION KEY

The ignition key completes the circuit flow when turned on .the AIR and tractive system de-energize when it turned OFF

5.6 FNR Switch

The F-N-R (Forward -Neutral-Reverse) switch is used to switch over from forward driving, Neutral condition and reverse mode. The signal from this switch is given to both the motor controller as well as to the Main Control unit for Motor operation and vehicle operating sequence as per rulebook.



Fig16. Ignition key and FNR Switch

5.7 Fuse

The fuse for protecting the Accumulator(s) tractive system circuit must have a rating lower than the maximum peak current of the isolation relays. Each of the electrical systems (both GLV and Tractive system) is protected by providing a fuse of the rating greater than the current rating of the electrical system. The current rating of a fuse is not greater than the peak current cut-off rating. The fuse is of automotive grade and it is instant blow type with 500 ms time delay.



Fig18. Fuse

3.Safety Systems		
3.01 TSAL Specifications		
Make / Model	LUMAX AUTOMOTIVE LTD.	
Color	Amber	
Flash Rate (Hz)	5	
Operating Voltage (V)	12	
IP Rating	IP67	
3.02 Ready to Drive Sound		
Make / Model	UNO MINDA	
Control Voltage (V)	12	
Sound intensity (dB) at 2m	70	
3.03 Fuse		
Fuse Location/Purpose	Current Rating (A)	Voltage Rating (V)
Battery	160	60
Controller	160	60
Type (Instant blow/Delay blow)	Instant Blow	
3.04 AIR		
Make / Model	Compag Automation	
Contact Current (A)	200	
Contact Voltage (V)	48	
Type (Normally Closed/Normally Open)	Normally Open	
3.05 Firewall		
Insulating layer thickness (mm)	3	
Insulating Material Make / Model	MG CHEMICALS	
3.06 Kill Switches		
Type	PUSH TO KILL AND ROTATE TO ENERGIZE	
Number of Kill Switches	2	
Working Voltage (V)	12	
3.07 Ignition Switch (If Applicable)		
Type (Switch,Key etc.,)	NO	
Working Voltage (V)	12	
3.08 HV Rated Cut-off Switch (If Applicable)		
Type (Switch,Key etc.,)	Switch	
Working Voltage (V)	48	
3.09 Reverse Alarm (If Applicable)		
Make/OEM	UNO MINDA	
Working Voltage (V)	12	
Location	Behind the Firewall	

Fig19. Specifications of all the GLV system components

Conclusion

Designing and implementing the E-Powertrain system for our SAE E-BAJA ATV has been both a challenging and rewarding journey. From the initial concept to the final assembly, we focused on creating a system that is not only efficient but also capable of withstanding the harsh demands of off-road conditions. Every electrical and electronic component used in this project was carefully chosen to meet safety standards and performance expectations.

Throughout the process, we faced several obstacles—from sourcing the right components and dealing with failures to navigating design complexities and system incompatibilities. However, each challenge pushed us to dig deeper, collaborate better, and improve our problem-solving approach. Using tools like FEA and CAE helped us fine-tune the system and make data-driven decisions that enhanced the reliability and efficiency of the final build.